What should we compare our results with?

What should be the baseline methods?

Baseline Methods

Choice of baselines

- Should I compare with non-peer-reviewed arXiv papers?
- Should I use their reported numbers or should I re-do the experiments?
  - Benefits of using their reported numbers:
    - Readers/Reviewers trust
  - Benefits of reimplementing the baselines
    - you can make sure all the ingredients that are *supposedly* the same are *actually* the same
    - you can use the same HP optimization procedure for your baselines
    - you can gain insight on the baselines and thereby compare and discuss differences more effectively
    - especially important when the improvement is incremental

Choice of baselines

- How broad should I do my literature study to find baselines?
  - Especially important with ML which has so many parallels in:
    - Control theory
    - Theoretical computer science
    - Statistics
    - Information theory
    - Data Mining
    - Traditional AI
    - …
  - A couple of years ago, I got a question from a person in another division: “do you compare with model identification techniques?”

Baseline Methods

Choice of baselines

- What dataset/issue/metric was the focus of the baseline?
  - Should we play their game, using their referee, in their home field?

We are not done yet!

Next Question: What is it in my model that brings this significant HP-optimized cross-validated improvements over the greatly chosen baselines?

Ablation Studies

Definition

Ablation studies refers to the removal/alteration of individual parts of the system to understand their importance to the desired outcome.

Essentially internal baselines!

Guidelines

- Have the same architecture as your baseline(s), if not, it's important to make sure to check each and every differences in the architecture!
- In the method section be clear about what you claim you bring on top of the baselines
  - As a parallel, in the experiments section make sure you do ablation studies
- Also be careful about the hidden differences:
  - e.g. duration of training
  - remember all hyperparameters…

Ablation Study

Fine-grained evaluation

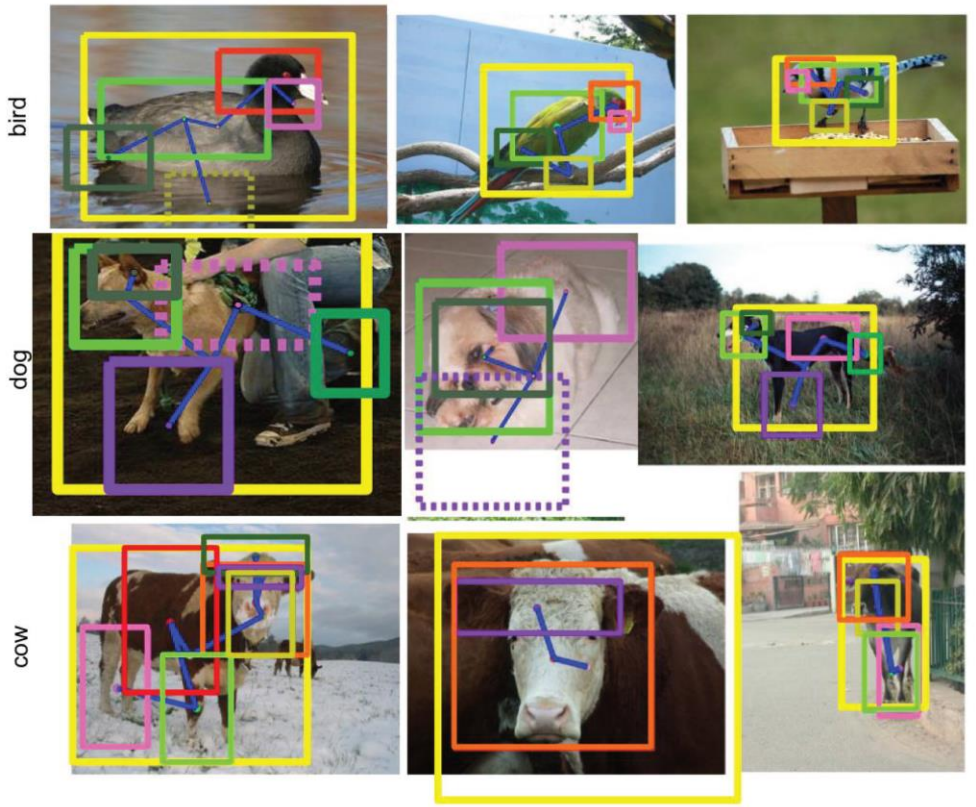
- Assume what each module should do, and then quantitatively or qualitatively measure that (beside the final performance measure)

- E.g., I add this module to the model be able to detect occluded objects.
  - Standard ablation study: remove that module and the detection performance drops
  - Example fine-grained ablation study: remove that module, check out the missing objects, they are occluded!

Ablation Study

Fine-grained evaluation

Show qualitative results (explicit modeling of occlusion)



[Azizpour&Laptev,"Object Detection Using Strongly-Supervised Deformable Part Models", ECCV 2012]